

How (not) to give a bad presentation

Slides by Rafael Bidarra and Petr Kellnhofer

Q4 2025/26



Realize that...

- A valuable message deserves to be well communicated (work/results).
- We can improve our presentation skills.
- It's awesome that a group of people offer us their time and attention!

Purpose of a presentation

- Make the audience intuitively grasp the essence of an idea (method, technique, solution: in the paper...)
- Excite and provoke them to read the paper.
- Make that process enjoyable.

Choosing the right content

1

Motivation
(20%)

2

Your key idea
(80%)

3

There is no 3

ACCEPT IT: YOU WILL NOT CONVEY ALL IDEAS IN THE PAPER...!

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(Simon Peyton Jones, *How to give a great research talk*)

Motivation



- Engage the audience before they start dozing or checking their phones.



- ... use short, very clear EXAMPLES (really!)

Motivation

- After the first 2-3 min, everybody should have answers to:
 - What is the problem?
 - Why does it matter?
 - What makes it challenging?
 - What's in it for me?

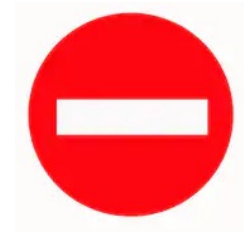
Heilmeier's Catechism

1. **What** are you trying to do?
 - Articulate your objectives using absolutely no jargon.
2. How is it done **today**, and what are the limits of **current** practice?
3. What's **new** in your approach and why do you think it will be successful?
4. **Who cares?** If you're successful, what difference will it make? What are the risks and the payoffs?
5. How much will it cost? How long will it take? What are the midterm and final "exams" to check for success?
(usually not so important for a paper)

This version is a formulation by Prof. Gordon Wetzstein (Stanford U)

For the original see <https://www.darpa.mil/work-with-us/heilmeier-catechism>

DO NOT:



- (start with) the outline of the talk.
 - Every talk has introduction and conclusion, we all know it
- Put ALL related work in a dry batch, **before your story**.
 - List of work without story is boring and useless.
- Maximize the amount of technical detail.

$$\begin{array}{c}
\frac{}{\Gamma \vdash k : \tau_k} \quad \frac{\Gamma \cup \{x : \tau\} \vdash e : \tau'}{\Gamma \vdash \lambda x. e : \tau \rightarrow \tau'} \quad \frac{\Gamma \vdash e_1 : \text{ST } \tau^\circ \tau \quad \Gamma \vdash e_2 : \tau \rightarrow \text{ST } \tau^\circ \tau'}{\Gamma \vdash e_1 \gg e_2 : \text{ST } \tau^\circ \tau'} \\
\\
\frac{\Gamma \vdash e : \tau}{\Gamma \vdash \text{returnST } e : \text{ST } \tau^\circ \tau} \quad \frac{\Gamma \vdash e : \tau}{\Gamma \vdash \text{newVar } e : \text{ST } \tau^\circ (\text{MutVar } \tau^\circ \tau)} \quad \frac{\Gamma \vdash e : \text{MutVar } \tau^\circ \tau}{\Gamma \vdash \text{readVar } e : \text{ST } \tau^\circ \tau} \\
\\
\frac{\Gamma \vdash e_1 : \text{MutVar } \tau^\circ \tau \quad \Gamma \vdash e_2 : \tau}{\Gamma \vdash \text{writeVar } e_1 e_2 : \text{ST } \tau^\circ \text{Unit}} \quad \frac{}{\Gamma \cup \{x : \forall \alpha_i. \tau\} \vdash x : \tau[\tau_i / \alpha_i]} \\
\\
\frac{\Gamma \vdash e : \tau' \rightarrow \tau \quad \Gamma \vdash e' : \tau'}{\Gamma \vdash e e' : \tau} \quad \frac{\Gamma \vdash e : \text{ST } \alpha^\circ \tau}{\Gamma \vdash \text{runST } e : \tau} \quad \alpha^\circ \notin FV(\Gamma, \tau) \\
\\
\frac{\forall j. \Gamma \cup \{x_i : \tau_i\}_i \vdash e_j : \tau_j \quad \Gamma \cup \{x_i : \forall \alpha_{j_i}. \tau_i\}_i \vdash e' : \tau'}{\Gamma \vdash \text{let } \{x_i = e_i\}_i \text{ in } e' : \tau'} \quad \alpha_{j_i} \in FV(\tau_i) - FV(\Gamma)
\end{array}$$

Figure 1. Typing Rules

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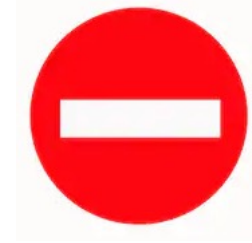
3

There is no 3

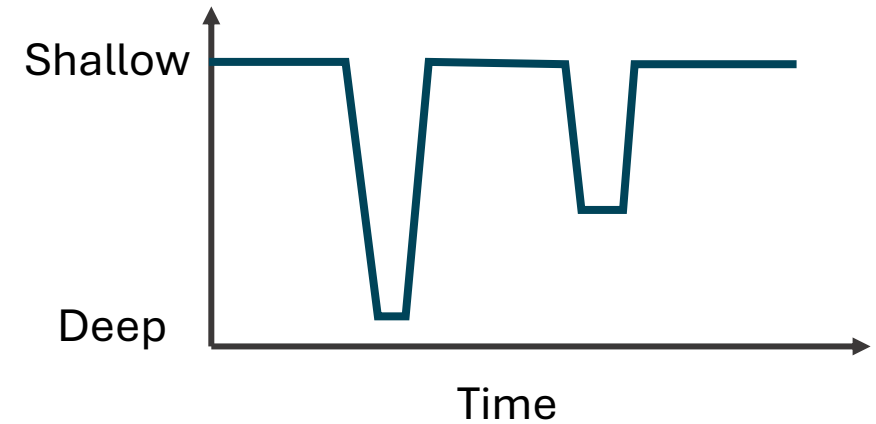
Key idea

- What is that **one thing** everyone should remember?
 - Be very specific, e.g., say explicitly:
"if you forget everything else, just keep this: ..."
- Visibly structure the whole talk around the key idea...
- Possibly in cycles, repeatedly coming back to it
- ...and prune the rest!

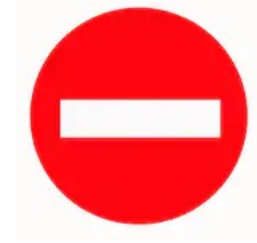
DO NOT:



- Give only a shallow overview.
 - **Better:** Explain the key idea instead of listing all components.
- Avoid any technical depth anywhere in the paper.
 - **Better:** Select cool part and go narrow and deep.
- Present the abstract model upfront.
 - **Better:** Use examples, build story.
- Try to cover equally all parts of the paper.
 - **Better:** Focus on what is important.

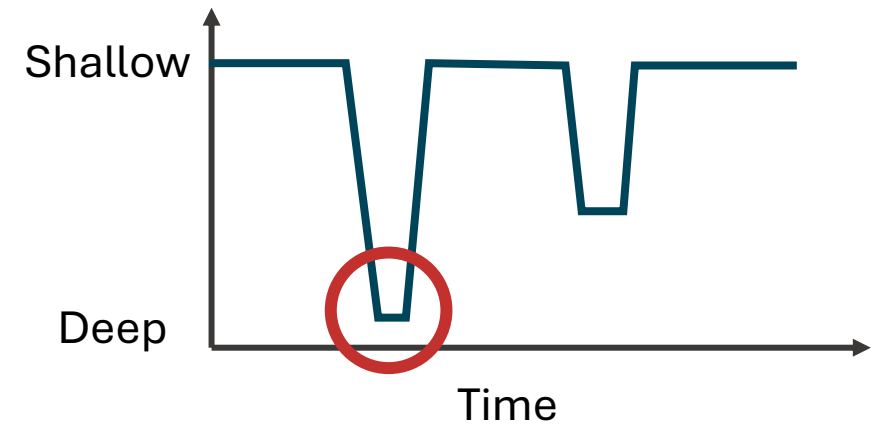


DO NOT:



- Gloss over equations.
 - Explain all variables (inputs and outputs),
 - Give an intuition of what the equation does and why. Where did it come from?

$$\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Slide 173

The benefits of being overly exhaustive
on every subject
you approach



Slides



Merging NCW with HSI

Man - Machine Interface considerations

Simplification

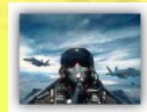
4th Dimension

Multi-task

Combat considerations

Easy Use

Military Interfaces



Cognition - Knowledge

Experience

Audio

Sense

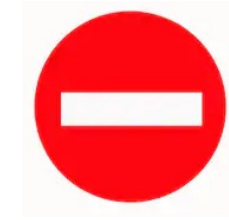
Visual

WISDOM

ACTION

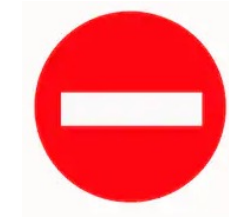


DO NOT:



- Use color abundantly...
 - ...e.g., in combination with some fancy 😊 characters 📎
- Use lots of bullets per slide...
 - ... and many indent levels...
 - ...to convey the hierarchy of your ideas
 - but show only...
 - ...one bullet...
- ... squeezing as much text as possible into each bullet, with complete sentences (including in-between side comments), even if the resulting bullet text wraps over several lines (for this, decrease figure and font sizes to fit in even more stuff).

DO NOT:



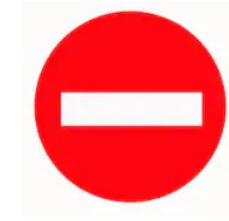
- Rely on a laser pointer.

Patrick Winston:

“There is a special circle in hell for those who use laser pointers”.

- Use graphical elements instead.

DO NOT:



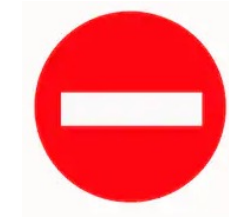
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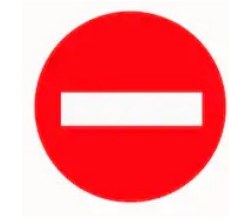
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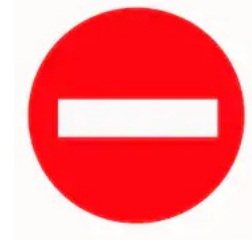
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Speaker

DO NOT



- Start by apologizing (for whatever...).
- Share some negative thought (delay, tech failure, weather complaint...).
- Assume the audience knows as much as you do.
- Look insistently at your screen, not at the audience.
- *Stand still, always behind the laptop...*
- *...and just read your slides (or notes).*

DO NOT



- Turn your back to the audience, to avoid any eye contact...
- ...and block the view to front row people.
- Keep an invariant, low tone of voice throughout the talk...
 - Better: Vary your pace, tone and volume for emphasis. Put breaks after important points to let them settle.
- Panic if you get lost.
 - Better: Take a short time, organize your thoughts and keep presenting.
- Clearly show your annoyance for being nervous.
 - Better: Always stay polite.

DO NOT



- Arrive last minute.
 - As a speaker come early and test your setup well before the lecture starts.
 - **Tip:** Bring your slides also on a flash drive in case a backup PC is needed.
- **Disregard any warning that you're over your time.**
 - Better: Plan ahead, jump to the important part and finish.
 - **Tip:** Practice time at home! – You can record yourself.

Conclusion



your paper
= the meal

your talk
= the meal
advertisement

do not confuse
the two



(Simon Peyton Jones, *How to give a great research talk*)

Additional materials

- Simon Peyton Jones, <https://www.microsoft.com/en-us/research/academic-program/give-great-research-talk/>
- Patrick Winston, <https://www.youtube.com/watch?v=RjbmPuhuFv0&list=PL240A9CB42552316A>
- Jim Blinn, <https://www.siggraph.org/s98/cfp/speakers/blinn.html>
- Frédo Durand, <http://people.csail.mit.edu/fredo/student.html>
- **Learning from other talks...!**